CONCEPT 45.1

Hormones and other signaling molecules bind to target receptors, triggering specific response pathways

A **hormone** (from the Greek *horman*, to excite) is a secreted molecule that circulates throughout the body and stimulates specific cells. Although a given hormone reaches all cells of the body, it only elicits a response—such as a change in metabolism—in specific *target cells*, those that have a receptor that binds the hormone specifically. Cells lacking a receptor for that hormone are unaffected.

Chemical signaling by hormones is the function of the **endocrine system**, one of the two basic systems for communication and regulation in the animal body. The other major communication and control system is the **nervous system**, a network of specialized cells—neurons—that transmit signals along dedicated pathways. These signals in turn regulate neurons, muscle cells, and endocrine cells. Because signaling by neurons can regulate the release of hormones, the nervous and endocrine systems often overlap in function.

As a background to our further exploration of the endocrine system, we'll begin with an overview of the diverse ways that animal cells use chemical signals to communicate.

Intercellular Information Flow

Communication between animal cells via secreted signals is often classified by two criteria: the type of secreting cell and the route taken by the signal in reaching its target. **Figure 45.2** illustrates five forms of signaling distinguished in this manner.

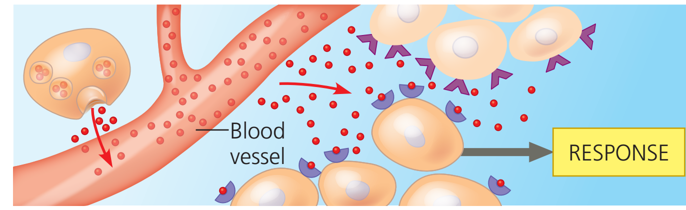
Endocrine Signaling

In endocrine signaling (see Figure 45.2a), hormones secreted into extracellular fluid by endocrine cells reach target cells via the bloodstream (or hemolymph). One function of endocrine signaling is to maintain homeostasis. Hormones regulate properties that include blood pressure and volume, energy metabolism and allocation, and solute concentrations in body fluids. Endocrine signaling also mediates responses to environmental stimuli, regulates growth and development, and triggers physical and behavioral changes underlying sexual maturity and reproduction (see Figure 45.1).

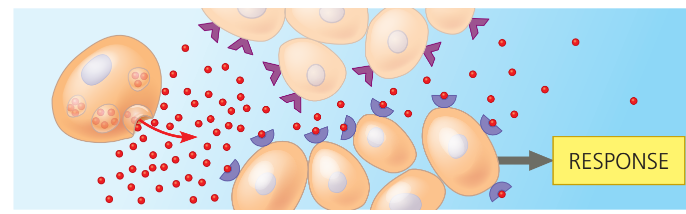
Paracrine and Autocrine Signaling

Many types of cells produce and secrete **local regulators**, molecules that act over short distances, reach their target cells solely by diffusion, and act on their target cells within seconds or even milliseconds. Local regulators play roles in many physiological processes, including blood pressure regulation, nervous system function, and reproduction.

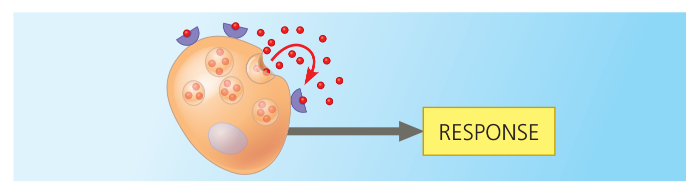
▼ **Figure 45.2 Intercellular communication by secreted molecules.** In each type of signaling, secreted molecules (•) bind to a specific receptor protein ( ) expressed by target cells. Some receptors are located inside cells, but for simplicity, here all are drawn on the cell surface.



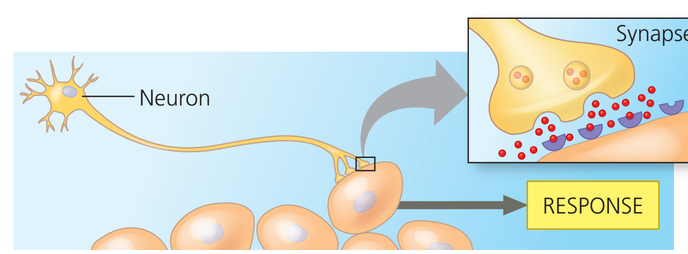
(a) In **endocrine signaling**, secreted molecules diffuse into the bloodstream and trigger responses in target cells anywhere in the body.



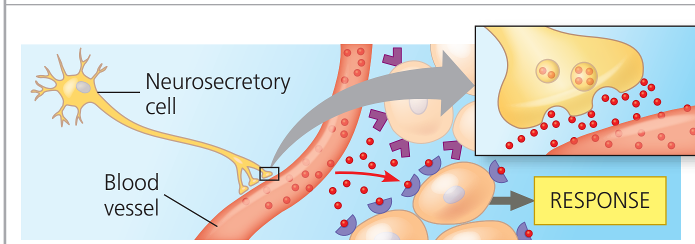
(b) In **paracrine signaling**, secreted molecules diffuse locally and trigger a response in neighboring cells.



(c) In **autocrine signaling**, secreted molecules diffuse locally and trigger a response in the cells that secrete them.



(d) In **synaptic signaling**, neurotransmitters diffuse across synapses and trigger responses in target tissues (neurons, muscles, or glands).



(e) In **neuroendocrine signaling**, neurohormones diffuse into the bloodstream and trigger responses in target cells anywhere in the body.

Depending on the target cell, signaling by local regulators is in general either paracrine or autocrine. In **paracrine** signaling (from the Greek *para*, to one side of), target cells